



**Auditable Plant-Level Digitalization Diagnosis under  
Clustered  
Observations: Measurement Audit, Virtual-Factory  
Benchmarking,  
and Guarded RIEC**

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Editor-in-Chief  
**Science Bulletin**

Dear Editor-in-Chief,

Please consider our manuscript entitled “**Auditable Plant-Level Digitalization Diagnosis under Clustered Observations: Measurement Audit, Virtual-Factory Benchmarking, and Guarded RIEC**” for publication in *Science Bulletin*.

**Core message.** This study is grounded in real cross-plant survey data from **COFCO** (one of the world's largest grain and oil industrial groups) and is aligned with industrial digitalization standardization through the **China Electronics Standardization Institute (CESI)**, China's national authority on related standards. Our goal is to make plant-level digitalization diagnosis **auditable and operationally robust** under clustered observations, where effective sample size can be far smaller than the raw count and rare but severe “tail” failures can mislead investment prioritization.

In large industrial groups, plant-level assessments are often assembled from diverse expert checklists and narrative reports. Two practical limitations arise in high-stakes use: (i) plants are nested within business units, inducing clustered observations and reducing the effective sample size; (ii) common selection rules emphasize average performance and may obscure rare but consequential generalization failures.

To address this, we propose **RIEC (Robustness-Integrated Evidence and Choice)**, an auditable end-to-end pipeline integrating three layers of evidence:

- (1) **Measurement evidence** from a data audit (missingness, ICC, and  $n_{\text{eff}}$ );
- (2) **Benchmarking evidence** from a calibrated virtual-factory world that generates controlled regimes for stress testing;
- (3) **Selection evidence** from transparent choice rules (CV, effective-sample BIC, and a hybrid RIEC criterion) with explicit tail-risk reporting.

We further introduce **Guarded-RIEC (RIEC-B)**, a lightweight “guard-and-retry” wrapper that triggers a conservative re-run when tail-risk signals indicate that the selected model is plausibly too complex for the audited information content. Across 50-seed benchmarks, RIEC-B remains stable in both central tendency and tail metrics (mean test MSE ~ 1.37-1.51;  $q_{99} \leq 2.28$ ), whereas BIC-like criteria and unguarded RIEC can exhibit heavy-tailed failures (with  $q_{99}$  up to ~300) despite similar medians.

For communication with non-technical stakeholders, we demonstrate an optional **SIRI16 anchor** that maps quantitative outputs to the Smart Industry Readiness Index (SIRI) vocabulary; the mapping design was co-developed with CESI to support standard-aligned industrial interpretation.

We provide Supplementary Information and a reproducibility bundle (code, figures, and runbook) to enable replication of the main results, including extended tail-risk analyses and a complete example of the SIRI16 anchor.

This submission is original and not under consideration elsewhere. All authors approve the submission and declare no competing interests. Generative AI tools were used only for English proofreading and final file preparation; all scientific content, analyses, and conclusions were produced and verified by the authors. We gratefully acknowledge **Cui Cheng** (a senior industry leader at COFCO) for discussions that inspired the initial motivation of this work.

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Thank you for your consideration.

Yours sincerely,

**Jincheng Li**  
on behalf of all authors

For Review Only

# Auditable Plant-Level Digitalization Diagnosis under Clustered Observations: Measurement Audit, Virtual-Factory Benchmarking, and Guarded RIEC

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## ABSTRACT

Plant-level digitalization diagnosis in large industrial groups is often assembled from diverse expert checklists and narrative reports, yielding inconsistent scores and opaque uncertainty. Because plants are nested within business units, observations are clustered and the effective sample size can be far smaller than the raw count, making downstream model selection unreliable—sometimes failing catastrophically in the tail. We propose **RIEC (Robustness-Integrated Evidence and Choice)**, an auditable pipeline that integrates (i) *measurement evidence* from a data audit (missingness, intraclass correlation coefficient (ICC), and effective sample size  $n_{\text{eff}}$ ), (ii) *benchmarking evidence* from controlled regimes generated by a calibrated virtual-factory world, and (iii) *selection evidence* from transparent choice rules that trade off predictive gain and complexity under small  $n_{\text{eff}}$ . We compare cross-validation (CV), an effective-sample BIC ( $\text{BIC}_{\text{eff}}$ ), and a hybrid RIEC criterion. We further introduce **Guarded-RIEC (RIEC-B)**, which detects and mitigates *tail catastrophes*: rare but severe generalization failures caused by selecting overly complex models under small  $n_{\text{eff}}$ . On a real group survey manifest (14 groups) used to calibrate the virtual world, the audit reports median  $\text{ICC} = 0.286$  and  $n_{\text{eff}} \approx 35$  in a sanity setting ( $n = 300$  synthetic rows). Across 50-seed benchmarks, CV and Guarded-RIEC remain stable (mean test  $\text{MSE} \approx 1.37\text{--}1.51$ ;  $q_{99} \leq 2.28$ ), while  $\text{BIC}_{\text{eff}}$  and unguarded RIEC exhibit heavy-tailed failures ( $q_{99}$  up to  $\sim 3 \times 10^2$ ) despite similar medians. Finally, we demonstrate an optional SIRI16 anchor that maps quantitative findings to a standard industry vocabulary for decision-makers.

## 1. Introduction

Plant-level diagnosis is a common first step in large-scale digital-transformation programs [1–4]. In large industrial groups, diagnosis outputs are often assembled from diverse expert checklists and narrative reports. This practice introduces two methodological difficulties that directly affect downstream decision-making:

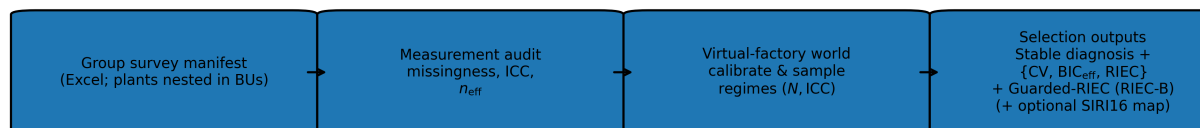
- **Inconsistent scoring and opaque uncertainty.** Different plants and experts use different rubrics; narrative reports are difficult to translate into a unified quantitative scale. As a result, uncertainty is rarely audited, and results are hard to reproduce across time or evaluators.
- **Clustered observations and small effective sample size.** Plants are nested within business units or professional companies (clusters). Within-cluster similarity induces non-i.i.d. data and can reduce the *effective* sample size far below the raw count, which can destabilize model selection and validation [5, 6].

These issues matter because diagnosis is not an end in itself: it is used to design digital-transformation roadmaps and allocate investment. A pipeline that is *not auditable* (unclear assumptions, unclear uncertainty, and unclear robustness) can lead to overconfident but fragile recommendations.

### 1.1. Why standard selection rules can be fragile under clustered observations

Two families of selection tools are commonly used in practice: information criteria (e.g., AIC/BIC) and predictive validation (e.g., CV). Both can be fragile in the clustered, low- $n_{\text{eff}}$  setting that is typical for group-level plant manifests.

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**Figure 1:** RIEC pipeline overview. A clustered group survey manifest is audited (missingness, ICC,  $n_{\text{eff}}$ ). A calibrated virtual-factory world generates controlled regimes (sample size  $N$ , ICC). Candidate model families are evaluated and selected by different rules; Guarded-RIEC (RIEC-B) mitigates heavy-tailed failures. Outputs can optionally be mapped to the SIRI16 vocabulary.

*Information criteria under “effective  $n$ ”.* Classical BIC is derived under i.i.d. sampling and asymptotic approximations [7]. When plants are nested within business units, the information content is closer to  $n_{\text{eff}}$  than to the raw  $n$ . A naive application of BIC can therefore under-penalize complexity, especially for flexible learners whose effective degrees of freedom are large. Using  $\text{BIC}_{\text{eff}}$  makes the dependence on  $n_{\text{eff}}$  explicit, but does not by itself guarantee tail stability.

*Cross-validation and the “average-risk” blind spot.* CV is widely used because it directly targets predictive performance [8]. However, in small or clustered samples, CV estimates can have high variance across folds and across random seeds. Moreover, CV is typically reported via the mean risk, which can obscure rare but severe failures. If a diagnosis informs large investments, the relevant question is not only “what is the expected error?” but also “how bad can it get in a rare run?”. This motivates combining CV with explicit tail-aware auditing.

*How RIEC addresses this gap.* RIEC is motivated by the observation that diagnosis must be robust and auditable, not only accurate on average. By auditing  $n_{\text{eff}}$ , benchmarking selection rules under controlled regimes, and reporting tail metrics (e.g.,  $q_{99}$ ), the pipeline makes hidden assumptions explicit and exposes failure modes early, before results are translated into managerial actions.

**Definition (RIEC).** We define **RIEC (Robustness-Integrated Evidence and Choice)** as an *auditable selection framework* that integrates three layers of evidence: (i) *measurement evidence* from a data audit (missingness, ICC,  $n_{\text{eff}}$ ); (ii) *benchmarking evidence* from regime-based stress tests in a calibrated virtual-factory world; and (iii) *selection evidence* from transparent decision rules (CV,  $\text{BIC}_{\text{eff}}$ , and RIEC-type criteria) with explicit tail-risk auditing under small  $n_{\text{eff}}$ . Operationally, RIEC outputs an audit report, a regime-wise benchmark report, and an auditable log of the final choice.

**Contribution.** We propose an end-to-end diagnosis pipeline (Fig. 1) and make four contributions: (1) a *measurement audit* tailored to clustered plant manifests (missingness, ICC,  $n_{\text{eff}}$ ); (2) a calibrated *virtual-factory world* that generates controlled regimes ( $N$ , ICC) for stress-testing selection rules; (3) a transparent comparison of CV,  $\text{BIC}_{\text{eff}}$ , and a hybrid RIEC criterion that trades off complexity and extrapolation gain under small  $n_{\text{eff}}$ ; and (4) **RIEC-B (Guarded-RIEC)**, a lightweight, auditable “guard rail” that reduces catastrophic tail failures without redesigning the whole pipeline.

## 2. Methods

### 2.1. Data: group survey manifest

We start from a plant-level survey manifest in which plants are grouped into business units (clusters). The manifest contains multiple indicators (features) and one or more target outcomes used for benchmarking model selection. In this paper, the manifest is used for (i) calibration of a virtual world and (ii) a sanity audit of measurement properties.

## 2.2. Measurement audit: missingness, ICC, and effective sample size

Let  $y_{ig}$  denote an observation for plant  $i$  in group  $g$ . A standard random-effects decomposition writes

$$y_{ig} = \mu + u_g + \varepsilon_{ig},$$

where  $u_g$  is a group-level effect and  $\varepsilon_{ig}$  is an idiosyncratic term. The intraclass correlation coefficient (ICC) is

$$\text{ICC} = \frac{\text{Var}(u_g)}{\text{Var}(u_g) + \text{Var}(\varepsilon_{ig})},$$

quantifying within-group similarity [5]. ICC induces a *design effect* and reduces the effective sample size  $n_{\text{eff}}$  relative to the raw  $n$  [6]. A common approximation is

$$n_{\text{eff}} \approx \frac{n}{1 + (\bar{m} - 1) \text{ICC}},$$

$$\bar{m} = \frac{n}{G},$$

where  $G$  is the number of groups and  $\bar{m}$  is the average group size. In our audit we report (i) indicator-wise missingness, (ii) ICC summaries across indicators, and (iii) an overall  $n_{\text{eff}}$  summary used downstream to regularize model selection.

## 2.3. Virtual-factory world: calibration and sampling regimes

To benchmark model-selection rules under controlled conditions, we build a calibrated *virtual-factory world*. The world is parameterized to match key measurement properties of the manifest. In our minimal calibration we match:

- **Cluster structure:** number of groups  $G$  and the empirical distribution of group sizes  $n_g$ .
- **Marginals:** indicator-wise means/variances after basic cleaning.
- **Clustering strength:** an ICC range consistent with the manifest audit.
- **Missingness pattern:** indicator-wise missingness rates (used for realism checks).

We then generate synthetic datasets under regimes defined by  $(N, \text{ICC})$  to stress-test selection rules. This is not a full causal simulator; rather, it is a controlled benchmark that makes uncertainty sources explicit and auditable.

## 2.4. Candidate set and model families

We compare candidate models in two structural families:

- **Pooled (no group effects):** ignores clusters and fits a global predictor.
- **GroupFE (group fixed effects):** includes group indicators to account for systematic group-level shifts.

Within each family we evaluate a small candidate set: linear regression, ridge regression, and gradient boosted decision trees (GBDT) [9]. Tree-based models offer flexibility but can be high-variance under small  $n_{\text{eff}}$ , motivating explicit tail-risk auditing.

## 2.5. Selection rules: CV, $\text{BIC}_{\text{eff}}$ , RIEC, and Guarded-RIEC (RIEC-B)

Let  $\hat{R}_{\text{cv}}(m)$  denote the  $K$ -fold cross-validated mean squared error (MSE) of model  $m$ , and let  $k(m)$  denote a complexity proxy.

CV. Select  $m$  minimizing  $\hat{R}_{\text{cv}}(m)$  [8].

$\text{BIC}_{\text{eff}}$ . For likelihood-based models, BIC is

$$\text{BIC}(m) = -2 \log L(m) + k(m) \log n.$$

We define an effective-sample variant by replacing  $n$  with  $n_{\text{eff}}$ :

$$\text{BIC}_{\text{eff}}(m) = -2 \log L(m) + k(m) \log n_{\text{eff}}.$$

Smaller  $n_{\text{eff}}$  increases the relative weight of complexity penalties.

RIEC. RIEC augments  $\text{BIC}_{\text{eff}}$  with an extrapolation-gain term. Let a baseline predictor (e.g., mean-only) have CV risk  $\hat{R}_{\text{cv}}^{\text{base}}$  and let

$$\text{XPE}(m) = \frac{\hat{R}_{\text{cv}}^{\text{base}}}{\hat{R}_{\text{cv}}(m)}.$$

so that  $\text{XPE}(m) > 1$  indicates an improvement over the baseline. A simple hybrid form is

$$C_{\lambda}(m) = \text{BIC}_{\text{eff}}(m) - \lambda(n_{\text{eff}}) \log(\text{XPE}(m)),$$

and RIEC selects  $m$  minimizing  $C_{\lambda}(m)$ , where  $\lambda(\cdot)$  controls the trade-off between structure/complexity and predictive gain. We use  $\lambda(n_{\text{eff}})$  as a monotone schedule that increases when  $n_{\text{eff}}$  is small.

**Guarded-RIEC (RIEC-B).** RIEC-B is an *operationally robust* variant of RIEC that adds an auditable “guard” to prevent tail catastrophes. Concretely:

1. Run RIEC to obtain an initial best model  $\hat{m}$  and diagnostics (e.g.,  $n_{\text{eff}}$ ,  $k(\hat{m})$ ,  $\hat{R}_{\text{cv}}(\hat{m})$ ,  $\text{XPE}(\hat{m})$ ).
2. If a **tail-risk trigger** fires (e.g., high complexity under small  $n_{\text{eff}}$  and unstable validation signals), re-run selection with a more conservative configuration (larger  $\lambda$  and/or a restricted candidate set that removes high-variance learners).
3. Log whether the guard triggered and which configuration was used (e.g., `guard_triggered`, `lambda_weight_used`) to make the decision auditable.

Examples of auditable trigger signals include: selecting a high-variance learner (e.g., GBDT) with high complexity when  $n_{\text{eff}}$  is small; unusually unstable CV estimates across folds; or extrapolation-gain signals that are inconsistent with the audited information content. Examples of conservative fallback actions include: increasing the penalty weight  $\lambda(n_{\text{eff}})$ ; restricting the candidate set to more stable model families (e.g., linear/ridge); and requiring a minimum validated improvement over a baseline before accepting added complexity. The exact trigger and fallback settings are pre-specified and reported in the Supplementary Information (SI).

## 2.6. Optional SIRI16 anchor

To support communication with non-technical stakeholders, we optionally map diagnosis outputs to the SIRI16 vocabulary (3 building blocks, 8 pillars, 16 dimensions) [10]. Concretely, the mapping outputs a ranked list of dimensions with the largest “impact” (gap×weight), producing an actionable top- $k$  focus list in standard industry terms. An illustrative SIRI profile and the resulting top- $k$  focus list are provided in the Supplementary Information (SI).

## 2.7. Evaluation protocol and tail-risk metrics

A practical diagnosis pipeline must be evaluated beyond “average” performance, because rare but severe failures can dominate operational risk. For each regime ( $N$ , ICC) in the virtual-factory world, we repeat the following loop over multiple random seeds: (i) sample a synthetic dataset, (ii) fit all candidate models, (iii) apply a selection rule (CV,  $\text{BIC}_{\text{eff}}$ , RIEC, or RIEC-B), and (iv) evaluate the selected model on an independent test set.

Let  $R_{\text{test}}^{(s)}$  denote the test MSE under seed  $s$ . We report the mean, the median, and a tail quantile across seeds:

$$\bar{R}_{\text{test}} = \frac{1}{S} \sum_{s=1}^S R_{\text{test}}^{(s)}, \quad \text{median}\left(\{R_{\text{test}}^{(s)}\}_{s=1}^S\right), \quad q_p\left(\{R_{\text{test}}^{(s)}\}_{s=1}^S\right),$$

where  $q_p$  is the  $p$ -quantile across seeds (e.g.,  $p = 0.99$ ). We interpret  $q_{99}$  as a *tail-risk* metric: “how bad can the diagnosis get in a rare run?”. In RIEC-B we additionally log whether the guard triggered and which conservative configuration was used, enabling post-hoc audit trails.

## 2.8. Audit artifacts: what is stored and why

RIEC is designed to be auditable end-to-end. Each run produces a set of artifacts that can be stored alongside the diagnosis report:

- **Manifest snapshot:** the exact input data (or a hash/manifest if privacy-constrained).



**Table 1**  
Measurement audit summary from the sanity run on the survey manifest.

| Metric                                         | Value | Interpretation                                                                   |
|------------------------------------------------|-------|----------------------------------------------------------------------------------|
| Number of groups (business units)              | 14    | Used to calibrate the virtual world.                                             |
| Synthetic training rows (sanity run)           | 300   | Synthetic rows sampled from the calibrated world for an end-to-end sanity check. |
| Median ICC across indicators                   | 0.286 | Moderate within-group correlation; violates i.i.d. assumptions if ignored.       |
| Overall effective sample size $n_{\text{eff}}$ | 35.3  | Information content $\approx$ 35 independent samples after clustering.           |

- **Measurement audit report:** missingness, ICC distribution, and  $n_{\text{eff}}$  summaries.
- **Regime configuration:** ( $N$ , ICC) settings and calibration metadata for the virtual-factory world.
- **Model score ledger:** per-model predictive risk and complexity summaries (CV risk, log-likelihood, criterion values).
- **Decision log:** the final selected model and (for RIEC-B) guard diagnostics such as `guard_triggered` and the penalty used.

These artifacts serve two purposes: (i) *reproducibility* (the same input should reproduce the same audited outputs), and (ii) *accountability* (stakeholders can inspect how the choice was made and whether a safety guard intervened).

### 3. Results

#### 3.1. Sanity audit on the manifest

Table 1 summarizes a measurement audit from an end-to-end sanity run. Although the sanity run uses  $n = 300$  synthetic training rows, the median ICC is 0.286 and the overall effective sample size is only  $n_{\text{eff}} \approx 35$ , indicating substantial clustering. This magnitude makes it plausible for flexible learners to overfit (or to pass CV on average) yet fail catastrophically in rare runs.

#### 3.2. Benchmark: mean performance hides heavy-tailed failures

We ran 50-seed benchmarks under four regimes  $(N, \text{ICC}) \in \{120, 240\} \times \{0.05, 0.25\}$ . For each regime and rule, we evaluate out-of-sample MSE on an independent test set and summarize both **central tendency** (mean/median) and **tail risk**. We define tail risk as the high quantile of test error across seeds, e.g.  $q_{99}$ .

Figure 2 reports the mean test MSE. The mean alone suggests that  $\text{BIC}_{\text{eff}}$  and RIEC can be much worse than CV/RIEC-B in some regimes, but it does not reveal *why*: whether performance is consistently poor or dominated by rare blow-ups.

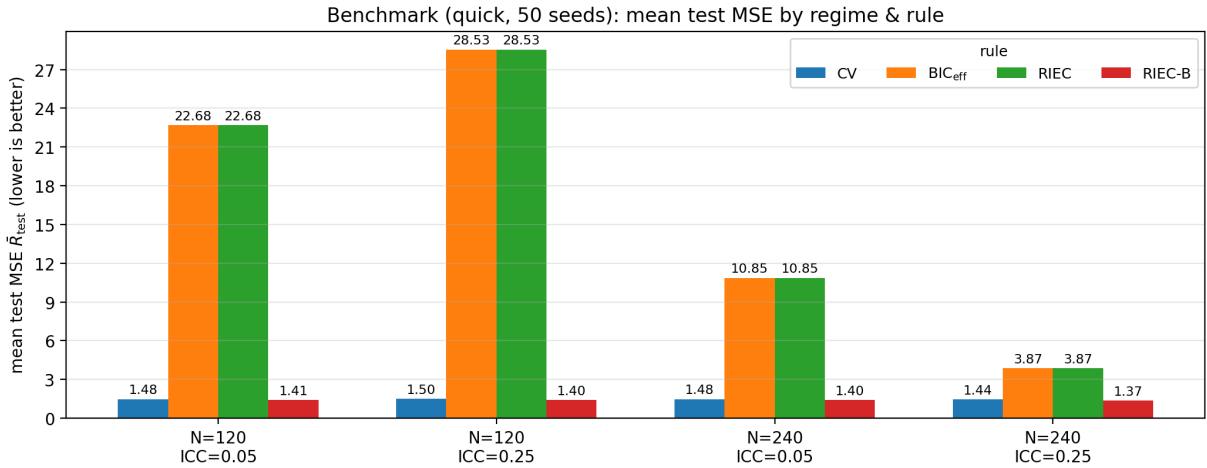
Figure 3 visualizes the seed-wise distribution (log scale). We observe a *tail-catastrophe* pattern:  $\text{BIC}_{\text{eff}}$  and ungarded RIEC have similar medians to CV, yet exhibit rare, extremely large test errors (heavy tails). This is precisely the failure mode that can be missed by mean-only reporting.

#### 3.3. Interpreting tail catastrophes and the role of the guard

Two empirical patterns are worth highlighting.

*Catastrophes are regime-dependent.* The largest tail risks appear in the more information-scarce regimes (smaller  $N$  and higher ICC). For example, Table 2 reports  $q_{99}$  near  $3.5 \times 10^2$  for  $N = 120$  and  $\text{ICC} = 0.25$ , while the same rules have smaller (but still non-trivial) tails when  $N = 240$  and  $\text{ICC} = 0.25$ . This is consistent with the audit perspective: higher ICC reduces  $n_{\text{eff}}$ , and lower  $n_{\text{eff}}$  amplifies the variance of complex learners.





**Figure 2:** Benchmark (50 seeds): mean test MSE  $\bar{R}_{\text{test}}$  by regime and selection rule. Values are annotated above bars for clarity. Lower is better.

**Table 2**

Tail-aware benchmark summary over 50 seeds in the calibrated virtual world (quick preset). Heavy-tailed failures inflate the mean and  $q_{99}$  for BIC<sub>eff</sub> and unguarded RIEC.

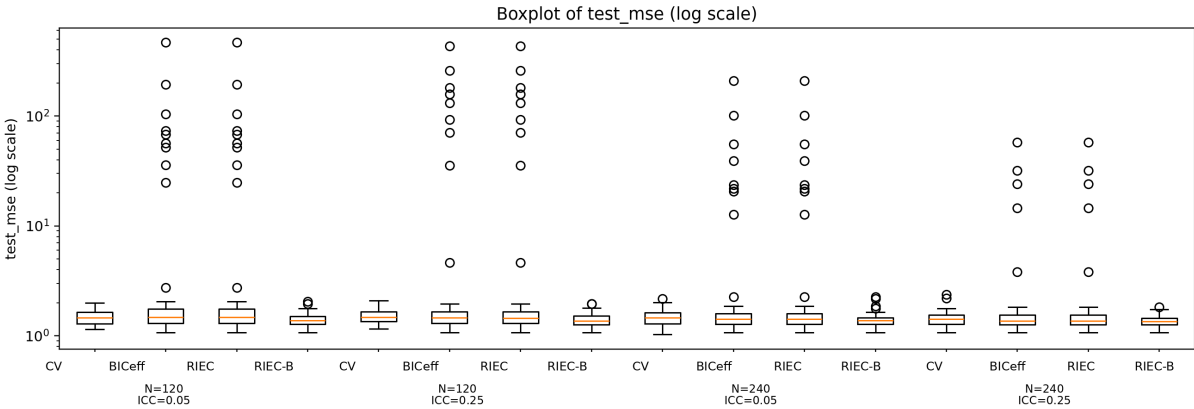
| Regime          | Rule               | Mean   | Median | $q_{99}$ |
|-----------------|--------------------|--------|--------|----------|
| N=120, ICC=0.05 | CV                 | 1.483  | 1.465  | 1.983    |
| N=120, ICC=0.05 | BIC <sub>eff</sub> | 22.678 | 1.482  | 332.468  |
| N=120, ICC=0.05 | RIEC               | 22.678 | 1.482  | 332.468  |
| N=120, ICC=0.05 | RIEC-B             | 1.409  | 1.374  | 2.000    |
| N=120, ICC=0.25 | CV                 | 1.505  | 1.472  | 2.071    |
| N=120, ICC=0.25 | BIC <sub>eff</sub> | 28.530 | 1.457  | 348.399  |
| N=120, ICC=0.25 | RIEC               | 28.530 | 1.450  | 348.399  |
| N=120, ICC=0.25 | RIEC-B             | 1.404  | 1.366  | 1.954    |
| N=240, ICC=0.05 | CV                 | 1.481  | 1.459  | 2.091    |
| N=240, ICC=0.05 | BIC <sub>eff</sub> | 10.852 | 1.415  | 156.698  |
| N=240, ICC=0.05 | RIEC               | 10.852 | 1.415  | 156.698  |
| N=240, ICC=0.05 | RIEC-B             | 1.404  | 1.372  | 2.211    |
| N=240, ICC=0.25 | CV                 | 1.443  | 1.413  | 2.280    |
| N=240, ICC=0.25 | BIC <sub>eff</sub> | 3.873  | 1.367  | 45.153   |
| N=240, ICC=0.25 | RIEC               | 3.873  | 1.368  | 45.153   |
| N=240, ICC=0.25 | RIEC-B             | 1.367  | 1.352  | 1.784    |

**BIC<sub>eff</sub> and unguarded RIEC can coincide.** In the quick benchmark reported here, BIC<sub>eff</sub> and unguarded RIEC produce identical summary statistics in Table 2. In this quick preset, the extrapolation-gain term often does not change the selected model relative to the information-criterion baseline, hence the identical summaries. This indicates that, under these settings, the hybrid criterion often makes the same final choice as the information-criterion baseline, and therefore inherits its tail failures. This motivates separating the *criterion design* (RIEC) from the *operational safety mechanism* (RIEC-B).

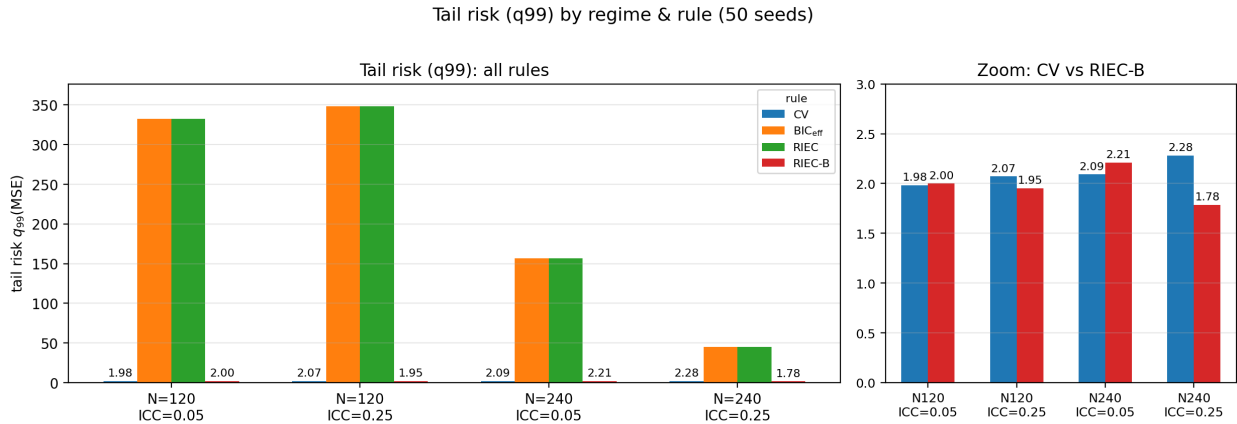
**RIEC-B is “RIEC + guard”.** We emphasize that RIEC-B is not a new model class; it is a wrapper that reduces operational risk:

$$\text{RIEC-B} = \text{RIEC} + \text{Guard}(\text{self-check followed by conservative re-run}).$$

The guard’s purpose is to catch situations where the selected model is plausibly too complex for the audited  $n_{\text{eff}}$  (or where validation signals are unstable), and to fall back to a conservative configuration. The value of this design is



**Figure 3:** Tail-catastrophe visualization (log scale). BIC<sub>eff</sub> and unguarded RIEC exhibit rare but extremely large test errors (heavy tails), even when medians are comparable.



**Figure 4:** Benchmark tail summary:  $q_{99}$  of test MSE by regime and rule (50 seeds). Left panel shows all rules; right panel zooms into the stable range to highlight CV vs RIEC-B. RIEC-B substantially reduces extreme failures relative to unguarded BIC<sub>eff</sub>/RIEC.

auditability: it is explicit when the guard intervenes, why it intervened, and what it changed (penalty weight, candidate restriction, etc.).

**Key finding: guarded selection reduces tail risk.** Across regimes, CV and Guarded-RIEC (RIEC-B) remain stable (mean test MSE  $\approx$  1.37–1.51;  $q_{99} \leq$  2.28), whereas BIC<sub>eff</sub> and unguarded RIEC show heavy-tailed failures with  $q_{99}$  up to  $\sim 3 \times 10^2$  despite similar medians (Table 2). Figure 4 summarizes  $q_{99}$  directly and includes a zoomed view to make the CV vs RIEC-B comparison visible.

#### 4. Discussion

**Why mean-only reporting is insufficient.** Our benchmarks show that mean test error can be dominated by rare catastrophic failures. For diagnosis pipelines that inform costly transformation decisions, reporting tail-aware metrics (e.g.,  $q_{95}$ ,  $q_{99}$ ) is more aligned with operational risk.

*RIEC vs. RIEC-B (why the guard matters).* RIEC provides a transparent, auditable bridge between information-criterion thinking ( $BIC_{\text{eff}}$ ) and predictive validation (CV) via an extrapolation-gain term. RIEC-B does not replace RIEC; it wraps it with a safety mechanism: when diagnostics indicate “too much complexity for too little  $n_{\text{eff}}$ ”, the guard forces a conservative re-run and records the decision. In industrial settings this mirrors standard safety engineering: a main controller (performance) complemented by an interlock (risk control).

*Practical implication.* If a diagnosis is used for prioritization and budgeting, a small probability of very large error can be unacceptable even if the median looks competitive. We recommend pairing any selection rule with (i) an  $n_{\text{eff}}$  audit and (ii) tail-aware benchmark summaries under plausible regimes.

*Deployment checklist.* For industrial use, we recommend the following minimal checklist:

- **Audit first:** report missingness, ICC distribution, and  $n_{\text{eff}}$  before fitting flexible models.
- **Stress-test under regimes:** benchmark candidate rules under a small set of plausible ( $N$ , ICC) regimes rather than a single “best guess”.
- **Report tails:** accompany mean/median with at least one tail metric (e.g.,  $q_{95}$  or  $q_{99}$  across seeds).
- **Store the decision log:** record the chosen model, criterion components, and (for RIEC-B) whether the guard intervened.
- **Translate to action:** if needed, map quantitative gaps to a standard vocabulary such as SIRI16 using a simple impact score (e.g.,  $\text{impact} = \text{gap} \times \text{weight}$ ) [10].

*Limitations and scope.* The virtual-factory world is calibrated to one manifest and represents a controlled benchmark, not a full causal model of factories. Future work should (i) expand the candidate set and regimes, (ii) validate on additional manifests, and (iii) study group-aware validation strategies (e.g., group-splitting CV) under realistic operational constraints.

## Data and code availability

A reproducibility bundle (scripts, figures, and runbook) accompanies this submission.

## CRedit authorship contribution statement

Jincheng Li: Conceptualization; Methodology; Formal analysis; Software; Investigation; Writing – original draft; Visualization. Xifeng Li: Resources; Data curation. Zhe Hao: Resources (WEF/SIRI context); Methodology (SIRI16 anchor). All authors: Writing – review & editing.

## Declaration of competing interest

The authors declare no competing interests.

## Declaration of generative AI and AI-assisted technologies

Generative AI tools were used for English-language proofreading and to assist in preparing the final submission files (e.g., formatting and consistency checks). All scientific content, analyses, and conclusions were produced and verified by the authors.

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# Supplementary Information: Auditable Plant-Level Digitalization Diagnosis under Clustered Observations

## 1 Notation, definitions, and key properties

### 1.1 Clustered data

We write observations as  $(x_{ig}, y_{ig})$  where  $i$  indexes plants within group (business unit)  $g \in \{1, \dots, G\}$ . Let  $n_g$  be the number of plants in group  $g$  and  $n = \sum_g n_g$ .

### 1.2 Intraclass correlation coefficient (ICC)

Under a random-intercept decomposition  $y_{ig} = \mu + u_g + \varepsilon_{ig}$ , the ICC is

$$\text{ICC} = \frac{\text{Var}(u_g)}{\text{Var}(u_g) + \text{Var}(\varepsilon_{ig})}.$$

**Interpretation:**  $\text{ICC} = 0$  indicates no within-group similarity; larger ICC indicates stronger clustering.

### 1.3 Effective sample size $n_{\text{eff}}$

Clustering inflates estimator variance via a *design effect*. A common approximation is

$$n_{\text{eff}} \approx \frac{n}{1 + (\bar{m} - 1) \text{ICC}}, \quad \bar{m} = \frac{n}{G},$$

with more refined variants using the empirical  $n_g$  distribution. In RIEC, we compute indicator-wise ICCs and report an overall  $n_{\text{eff}}$  summary used for conservative regularization.

**Property 1 (i.i.d. limit).** If  $\text{ICC} = 0$ , then  $n_{\text{eff}} \approx n$  and all “effective-sample” adjustments reduce to their standard i.i.d. counterparts.

### 1.4 Model selection rules

Let  $\hat{R}_{\text{cv}}(m)$  denote the  $K$ -fold cross-validated risk (MSE) of model  $m$ ; let  $k(m)$  denote a complexity proxy (degrees of freedom / parameter count / tree depth proxy, depending on model class).

**CV.** Select  $m$  minimizing  $\hat{R}_{\text{cv}}(m)$ .

**BIC<sub>eff</sub>.** For likelihood-based models, BIC is

$$\text{BIC}(m) = -2 \log L(m) + k(m) \log n.$$

We define an effective-sample variant by replacing  $n$  with  $n_{\text{eff}}$ :

$$\text{BIC}_{\text{eff}}(m) = -2 \log L(m) + k(m) \log n_{\text{eff}}.$$

**Interpretation:** smaller  $n_{\text{eff}}$  increases the relative weight of complexity penalties.

**RIEC.** RIEC augments  $\text{BIC}_{\text{eff}}$  with an extrapolation gain term. Define a baseline predictor (e.g., mean-only) with CV risk  $\hat{R}_{\text{cv}}^{\text{base}}$  and the model's CV risk  $\hat{R}_{\text{cv}}(m)$ . Define

$$\text{XPE}(m) = \frac{\hat{R}_{\text{cv}}^{\text{base}}}{\hat{R}_{\text{cv}}(m)}.$$

Then a simple hybrid form is

$$C_{\lambda}(m) = \text{BIC}_{\text{eff}}(m) - \lambda(n_{\text{eff}}) \log(\text{XPE}(m)),$$

and RIEC selects  $m$  minimizing  $C_{\lambda}(m)$ , where  $\lambda(\cdot)$  controls the trade-off between structure/complexity and predictive gain.

**Guarded-RIEC (RIEC-B).** RIEC-B adds an auditable guard:

1. Run RIEC to obtain an initial best model  $\hat{m}$  and diagnostics (e.g.,  $n_{\text{eff}}$ ,  $k(\hat{m})$ ,  $\hat{R}_{\text{cv}}(\hat{m})$ ,  $\text{XPE}(\hat{m})$ ).
2. If a tail-risk trigger fires (e.g., high complexity under small  $n_{\text{eff}}$  and unstable validation signals), re-run selection with a more conservative penalty (larger  $\lambda$ ) and/or a restricted candidate set.
3. Log whether the guard triggered and which penalty was used (`guard_triggered`, `lambda_weight_used`).

**Property 2 (tail-risk mitigation by conservatism).** Empirically, in our 50-seed benchmarks the guard substantially reduces extreme quantiles ( $q_{99}$ ) relative to unguarded BIC/RIEC, even when medians are similar. This supports the use of tail-aware auditing for high-stakes diagnosis pipelines.

## 2 Additional figures

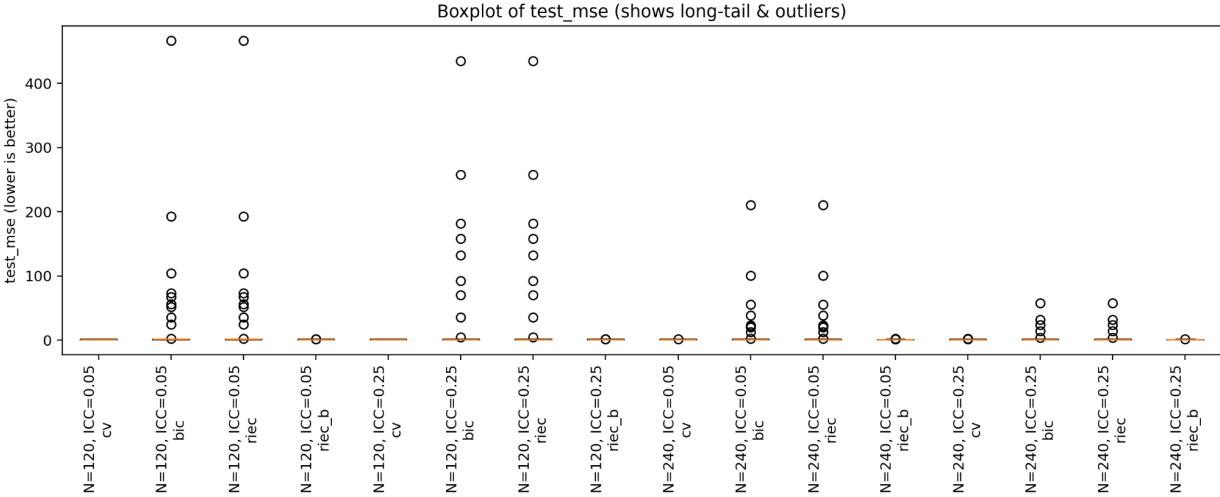


Figure S1: Seed-wise test MSE distributions (linear scale) illustrating long-tail failures. BIC and unguarded RIEC show rare but severe outliers under some regimes, while Guarded-RIEC (RIEC-B) remains stable.

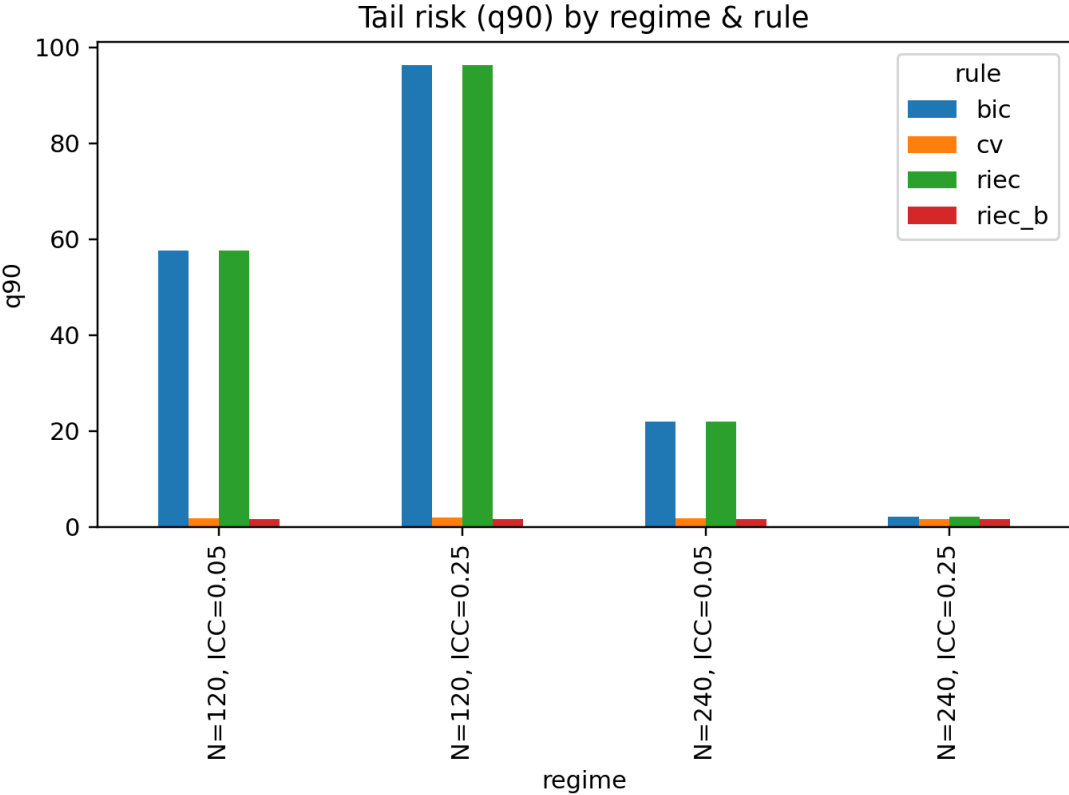


Figure S2: Tail-risk summary at the 90th percentile ( $q_{90}$ ) by regime and selection rule.



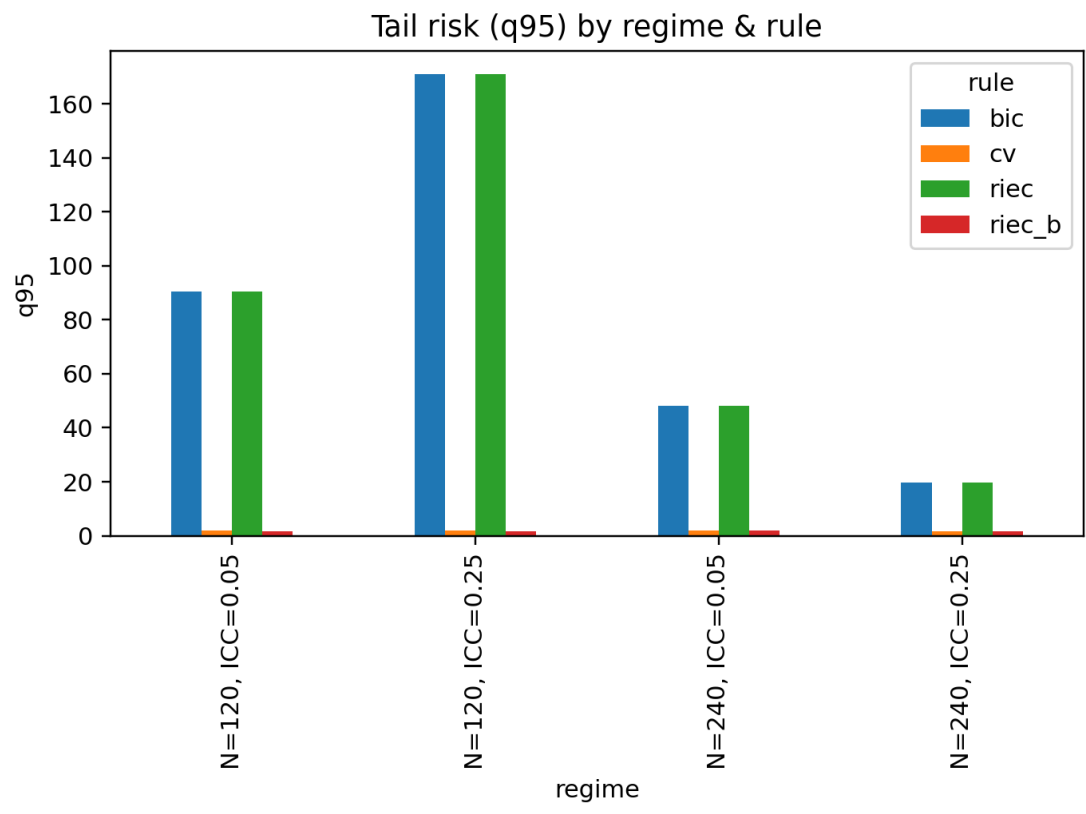


Figure S3: Tail-risk summary at the 95th percentile ( $q_{95}$ ) by regime and selection rule.

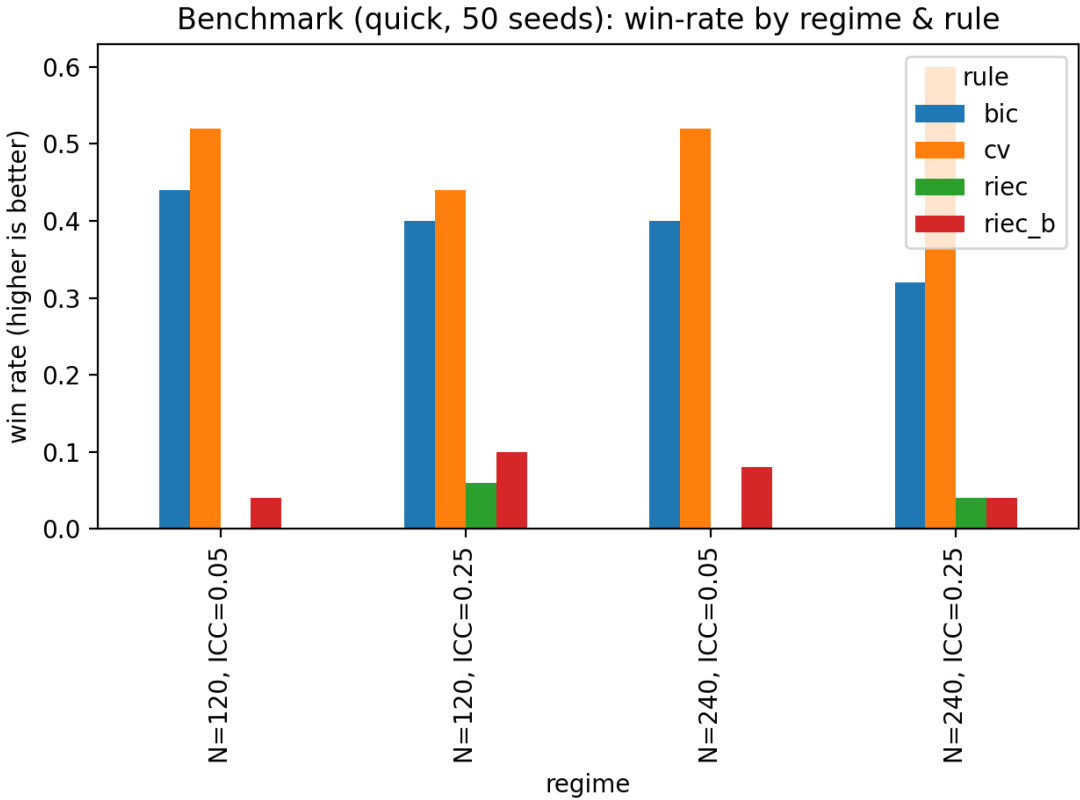


Figure S4: Win rate by regime and selection rule in the quick benchmark (50 seeds). Win rate is the fraction of seeds where a rule attains the lowest test MSE within a regime.

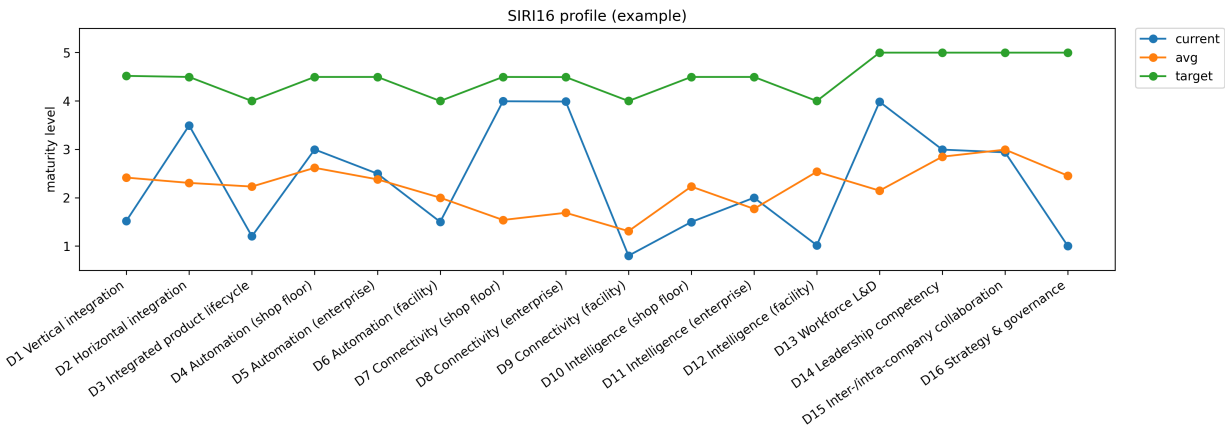


Figure S5: Example SIRI16 maturity profile mapped from an assessment export. We show a current profile, a portfolio-average profile, and an aspirational/benchmark target profile for gap analysis.

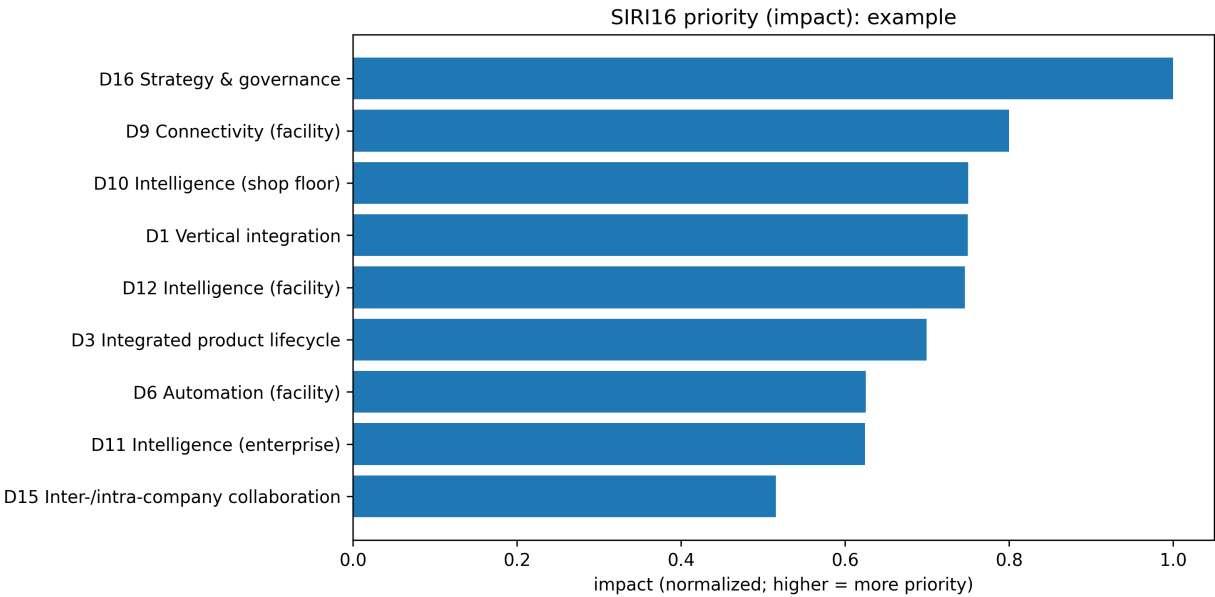


Figure S6: Example SIRI16 prioritization output. Dimensions are ranked by a normalized impact score based on maturity gaps (here: target–current under equal weights).

### 3 Supplementary tables

- `tables/Table2_benchmark_tail_summary.csv`: Seed-level benchmark summary (mean, median,  $q_{99}$ ) for each regime and rule.

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| regime    | rule   | mean   | median | q99     |
|-----------|--------|--------|--------|---------|
| N=120, IC | bic    | 22.678 | 1.482  | 332.468 |
| N=120, IC | cv     | 1.483  | 1.465  | 1.983   |
| N=120, IC | riec   | 22.678 | 1.482  | 332.468 |
| N=120, IC | riec_b | 1.409  | 1.374  | 2.000   |
| N=120, IC | bic    | 28.530 | 1.457  | 348.399 |
| N=120, IC | cv     | 1.505  | 1.472  | 2.071   |
| N=120, IC | riec   | 28.530 | 1.450  | 348.399 |
| N=120, IC | riec_b | 1.404  | 1.366  | 1.954   |
| N=240, IC | bic    | 10.852 | 1.415  | 156.698 |
| N=240, IC | cv     | 1.481  | 1.459  | 2.091   |
| N=240, IC | riec   | 10.852 | 1.415  | 156.698 |
| N=240, IC | riec_b | 1.404  | 1.372  | 2.211   |
| N=240, IC | bic    | 3.873  | 1.367  | 45.153  |
| N=240, IC | cv     | 1.443  | 1.413  | 2.280   |
| N=240, IC | riec   | 3.873  | 1.368  | 45.153  |
| N=240, IC | riec_b | 1.367  | 1.352  | 1.784   |

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